

# SOLUTION

## 1: WATER JUG PROBLEM

### Solution

The Water Jug Problem is solved using state-space search, where each state represents the amount of water in the two jugs. The objective is to obtain exactly 2 gallons of water in the 4-gallon jug by performing valid operations.

### Initial State

- (0,0) – Both jugs are empty.

### Goal State

- (2,0) – The 4-gallon jug contains exactly 2 gallons.

### Allowed Operations

- Fill a jug completely.
- Empty a jug.
- Pour water from one jug to another until one jug is full or the other is empty.

### Solution steps:

| STEP | OPERATION                                              | 4 – GALLON | 3 – GALLON |
|------|--------------------------------------------------------|------------|------------|
| 1    | Fill the 3-gallon jug                                  | 0          | 3          |
| 2    | Pour into the 4-gallon jug                             | 3          | 0          |
| 3    | Fill the 3-gallon jug again                            | 3          | 3          |
| 4    | Pour water into the 4-gallon jug until it becomes full | 4          | 2          |
| 5    | Empty the 4-gallon jug                                 | 0          | 2          |
| 6    | Transfer the remaining 2 gallons to the 4-gallon jug   | 2          | 0          |

- **Goal Achieved:** The 4-gallon jug contains exactly 2 gallons.

### **Result**

The goal state **(2,0)** is reached successfully.

### **Conclusion**

This problem demonstrates how an AI agent explores different states systematically to achieve a goal using an uninformed search strategy.

## **2: MARS ROVER**

### **Solution**

The Mars Rover is an autonomous intelligent agent designed to explore Mars without direct human control.

### **Percepts**

- Camera images
- Rock and soil samples
- Obstacle information
- Temperature
- Battery level
- Position and terrain

### **Operating Environment**

- Partially observable
- Dynamic
- Sequential
- Continuous
- Unknown
- Remote environment

### **Actions**

- Move forward and backward

- Turn left and right
- Capture images
- Collect rock samples
- Analyze soil
- Avoid obstacles
- Send data to Earth

### **Performance Measure**

- Collect maximum scientific data
- Navigate safely
- Minimize energy consumption
- Complete mission successfully

### **Suitable Agent**

Utility-Based Agent

### **Justification**

A utility-based agent evaluates several possible actions and selects the one that maximizes mission success while minimizing energy consumption and risk. It performs efficiently in uncertain and changing environments.

### **Conclusion**

The Mars Rover is an excellent example of an intelligent autonomous agent that uses AI to make decisions in real time.

## **3: 8-QUEENS PROBLEM**

### **Solution**

The 8-Queens Problem is a Constraint Satisfaction Problem (CSP). The objective is to place eight queens on an 8×8 chessboard so that no queen attacks another queen.

### **Problem Formulation**

#### **Initial State**

- Empty chessboard

### **State Space**

- All possible arrangements of queens

### **Operators**

- Place a queen
- Remove a queen
- Move a queen

### **Goal Test**

- Eight queens are placed.
- No two queens share the same row, column, or diagonal.

### **Path Cost**

- Each queen placement is treated as one step.

### **Search Strategy**

- Backtracking Search
- Depth-First Search (DFS)

### **Advantages**

- Eliminates invalid configurations early.
- Efficiently explores the search space.
- Guarantees a valid solution if one exists.

### **Conclusion**

The 8-Queens Problem demonstrates AI search techniques and constraint satisfaction methods used in scheduling, planning, and optimization.

## **4: OLA CAB BOOKING**

### **Solution**

The OLA Cab Booking application uses an intelligent agent to recommend the best cab according to the customer's travel needs.

### **Agent Type**

Goal-Based Agent

## **Working Process**

1. User enters pickup location.
2. User enters destination.
3. System checks available cabs.
4. User selects a preferred cab type.
5. System calculates fare and estimated arrival time.
6. Booking is confirmed.

## **Pseudocode**

START

Input Pickup Location

Input Destination

Display Available Cabs

Select Cab Type

Calculate Fare

Confirm Booking

END

## **Advantages**

- Fast booking process.
- Suggests suitable cab options.
- Improves customer satisfaction.
- Reduces waiting time.

## **Conclusion**

A goal-based agent makes decisions based on the user's objective and selects the most appropriate cab to complete the journey efficiently.

## 5: UNIFORM COST SEARCH (UCS) – SOLUTION

### Solution

Uniform Cost Search (UCS) is an uninformed search algorithm that always expands the node with the lowest cumulative path cost. It guarantees the optimal solution when all edge costs are non-negative.

### STEP 1: GRAPH DETAILS:

Edges and Costs:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

### STEP 2: UCS SEARCH PROCESS:

**Step**   **Expanded Node**   **Path Cost**   **Frontier (Queue)**

**Path Cost**

$S \rightarrow A \rightarrow C \rightarrow G$  4

$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$  6

$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$  10

$S \rightarrow G$  12

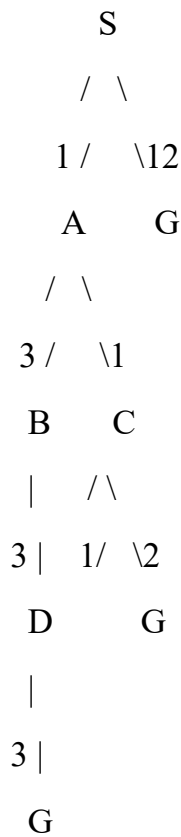
### STEP 3: LEAST-COST PATH

The optimal path found by UCS is:

$$S \rightarrow A \rightarrow C \rightarrow G$$
**TOTAL COST**

$$1 + 1 + 2 = 4$$

### STEP 4: SEARCH TREE:



**Optimal Path: S  $\rightarrow$  A  $\rightarrow$  C  $\rightarrow$  G**

**Total Cost = 4**

## Conclusion

Uniform Cost Search is an efficient algorithm for solving shortest-path problems because it always selects the path with the minimum accumulated cost.